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Parishruthi Ganesh

Computer Vision (CS) Ph.D. Student

First-author preprints in video understanding and model evaluation, submitted to WACV and AAAI. Research in weakly-supervised violence detection, human pose and interaction representations, person detection and tracking, object detection, and what video benchmarks actually measure. Past experience includes Auburn University and Emids Technologies, and seeking computer vision research internships.

Education

Ph.D. in Computer Science and Software Engineering — **Aug 2025 – Present**
Auburn University · Focus: Computer Vision, Video Understanding, Object Detection

Master of Science in Computer Science and Software Engineering — **Aug 2023 – Aug 2025**
Auburn University · GPA 3.82 / 4.00

Bachelor of Engineering in Computer Science — **Aug 2017 – Aug 2021**
Visvesvaraya Technological University

Skills

Research: Video Understanding, Human Activity & Violence Detection, Object Detection, Person Detection & Tracking, Pose & Interaction Representations, Weak Supervision, Benchmark Validity, Robustness & Calibration
Languages: Python, Java, C++, SQL, JavaScript
Frameworks & Tools: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, Deep Graph Library (DGL), Docker, AWS, Weights & Biases

Computer Vision & Machine Learning Publications

- What Do Interaction Representations Actually Measure? Pre-Event Separability in Weakly-Supervised Violence Detection; **Submitted to WACV**; arXiv:2608.27879
- Selecting Open-Weight Language Models for Zero-Shot Intent Classification: A Systematic Evaluation of 41 Models; **Submitted to AAAI**; arXiv:2607.27421

Research & Machine Learning Experience

Graduate Research & Teaching Assistant, Auburn University — **Jan 2024 – Present**
- Research across video understanding and applied machine learning, alongside a retrieval pipeline over roughly 25,000 research papers and a digital-engineering data platform.
- Deliver hands-on Python and MATLAB lab sessions to undergraduates, supporting programming proficiency through mentoring and assignment guidance.

Associate Software Developer, Emids Technologies Pvt Ltd — **Dec 2021 – Jul 2023**
- Led development of a CRM platform covering marketing, product, customer service and sales operations; built the Leads, Contacts, Email API, Reports and Accounts modules.
- Designed Spring Boot services and RESTful APIs, and built the server-side layer with JSP, JSTL and Servlets.

Selected Machine Learning Projects

SplitShield *(Proof of Possible 2026; GitHub; Demo; Video)*
- Finds duplicate and near-duplicate images leaking across dataset splits with perceptual hashing and MobileNetV3-Small embeddings, repairs the split, and reports how far accuracy moves.

Circular Trade Fraud Detection with Graph Learning *(Research project; GitHub)*
- Detects circular trading rings in invoice data with graph cycle detection, Node2Vec embeddings and DBSCAN.

Sentinel Memory *(CockroachDB × AWS — Build with Agentic Memory; GitHub; Demo; Video)*
- An incident-response agent whose memory is a transactional database rather than a context window.

LabGuard AI *(All Things Agentic Hackathon; GitHub; Demo; Video)*
- An autonomous research agent that challenges a scientific claim and runs the experiments that could disprove it.

AstraNova *(Build with Gemini XPRIZE; Demo; Video)*
- An intraday options copilot combining live market data, model-scored signals and broker execution.

ThinkTrace AI *(SPEED August AI Challenge; GitHub; Demo; Video)*
- Reads each student's written reasoning rather than only the final answer, and groups a class by shared misconception.

Multi-Class Sentiment Analysis with BERT *(Coursework; GitHub)*
- Fine-tuned bert-base-uncased on the SMILE annotation dataset as a study of transformer transfer learning.

Leadership & Service

- Treasurer, The Society for Responsible Artificial Intelligence (December 2024 – Present).